

Project Report: Hotel Booking Cancellation Analysis Using Excel

1. Project Overview

This project focuses on analyzing hotel booking data to understand cancellation behavior, guest preferences, seasonal demand, and key factors influencing booking outcomes. Using Excel, Pivot Tables, and an interactive dashboard, the analysis highlights patterns in customer behavior and provides insights that can help hotels reduce cancellations and improve revenue forecasting.

2. Objectives of the Project

The primary goals of the analysis were:

1. To study the **rate of booking cancellations** across different customer types.
2. To examine **seasonal trends** in bookings and cancellations.
3. To understand how factors like **lead time, location, guest type, room type**, and **number of special requests** influence cancellations.
4. To create a **dynamic dashboard** that visualizes cancellations using slicers and filters.

3. Tools & Techniques Used

Excel Tools

- Pivot Tables
- Pivot Charts
- Slicers & Timelines
- Data Cleaning (removing blanks, fixing inconsistencies)
- Interactive Dashboard Creation

4. Data Cleaning & Preprocessing

Performed the following cleaning steps:

- Removed duplicate rows
- Standardized categorical values (Guest Type, Hotel Type, Meal Type)
- Converted booking date into **Year, Month, and Week**
- Created new calculated fields (e.g., Total Nights, Is_Cancelled flag)
- Handled missing values in fields like Country & Agent

5. Pivot Table Analysis

Several pivot tables were created to analyze booking and cancellation trends:-

a) Cancellation Rate by Guest Type

- “Transient” guest type showed the **highest cancellation rate**.
- “Group” bookings were more stable with fewer cancellations.

b) Month-wise Booking & Cancellation Trends

- High cancellations observed in **summer months** and year-end periods.
- Booking demand spikes during holiday seasons.

c) Lead Time vs Cancellation

- Longer lead times (early bookings) had a **significantly higher cancellation rate**, indicating uncertain plans.

d) Room Type & Market Segment Analysis

- Certain room types saw more cancellations due to pricing differences.
- Online Travel Agents contributed the maximum number of cancellations.

e) Special Requests

- Guests with **more special requests** had a lower likelihood of cancellation — indicating higher commitment.

6. Dashboard Components

The dashboard includes multiple visual insights:

• KPI Cards

- Total Bookings
- Total Cancellations
- Cancellation Rate
- Most Active Guest Type
- Peak Booking Months

• Monthly Trend Chart

Shows how cancellations vary by month.

• Guest Type Breakdown

Bar chart comparing confirmed vs cancelled bookings for each type.

- **Location-Based Analysis**

Shows which regions contribute more cancellations.

- **Lead Time Distribution**

Highlights how the booking lead time correlates with cancellations.

- **Dynamic Slicers**

Dashboard includes filters for:

- Guest Type
- Arrival Year
- Meal Type
- Hotel Type
- Market Segment

7. Key Insights & Findings

- **Longer lead times correlate strongly with cancellations.**
- **Transient and Online Travel Agent bookings** show the highest cancellation counts.
- **City Hotels** have higher booking volumes but also higher cancellation rates than Resort Hotels.
- **Summer and December** show peak cancellations due to high demand and flexible travel plans.
- **Guests with special requests or longer stays** tend to cancel less frequently.
- **Family guests (Family or Couple segments)** show more stable booking behavior.

8. Conclusion

This project demonstrates how data analysis and dashboarding can be used to understand cancellation behavior and forecast future demand. Through pivot tables and visual insights, we identified key patterns impacting revenue and operational planning.

The project reflects strong skills in:

- Data cleaning
- Pivot table summarization
- Dashboard storytelling
- Understanding customer and business behavior
- Analytical interpretation of booking trends

These insights can be used by hotels to reduce cancellations, optimize pricing strategies, and improve revenue management.